

K. SRINIVAS KARTHIK

AI & ML Engineering Student

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AREAS OF INTEREST

Machine Learning | Deep Learning | Computer Vision | Artificial Intelligence | Quantum Computing (Foundations) | Data Science

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML) — GITAM University, Visakhapatnam 2025 – 2029

CGPA: 9.9 / 10.0 | Full Merit Scholarship

- Relevant Coursework: Programming in C, Python, Data Structures & Algorithms (basics), Basics of Agentic AI

Class XII — AP Board 2025

Percentage: 96.7%

Class X — CBSE 2023

Percentage: 84%

PROFESSIONAL EXPERIENCE & LEADERSHIP

Full Stack Developer Intern — Inframiq Solutions Pvt. Ltd. (Remote) Jul 2026 – Present

- Co-building a product end-to-end as one of two developers, owning implementation across the frontend, backend, and database layers from initial architecture onward.
- Currently supporting the team through the product's pitching phase ahead of public launch.

Head of Operations — Meta Developer Communities (MDC), GITAM Visakhapatnam Jul 2026 – Present

- Lead operations for a student tech community spanning seven domains — Competitive Programming, WebArcs, DataVerse, Design, Photography, Content, and PR — under the Directorate of Student Life.
- Ran the AY 26–27 recruitment cycle, coordinating candidate communications across WhatsApp and email.
- Built a Google Apps Script automation for new-member onboarding across all seven domains, generating Letters of Selection and routing welcome emails and WhatsApp group placement by domain.
- Maintain operational infrastructure including Apps Script certificate pipelines and Notion-based ops boards.
- Drive content and infrastructure for Meta Code Quest, a 10-chapter monthly coding hackathon series — producing event reports, social announcements, and certificate automation for each chapter.
- Produced promotional and operational content for the Feature Forge ML workshop (~139 attendees), including a promotional reel and certificate mailer.

PROJECTS

Vision-Guided Robotic Arm — Python · OpenCV · MediaPipe · PyTorch · YOLOv8 · Stable-Baselines3 · Gymnasium · MATLAB Simulink/Simscape

- Led the ML/Computer Vision side of a hand-gesture-controlled robotic arm for the Robotics Club, iterating the design from a 5-finger anthropomorphic hand to a 2-finger parallel gripper for reliability.
- Built a real-time hand-tracking pipeline in OpenCV and MediaPipe with a scale-invariant, zero-calibration angle-based aperture calculation and an EMA smoothing filter to stabilize gripper commands.
- Designed a simulation-first control architecture bridging Python (perception/RL) and MATLAB Simulink/Simscape (physics) over TCP sockets using a custom JSON protocol, and optimized video capture with a threaded CameraThread implementation for improved real-time FPS on an RTX 4070 laptop GPU.

IoT Smart Humidifier — ESP32 · Blynk IoT · C++

- Engineered an ESP32-based automated humidifier with real-time DHT11 sensor monitoring and threshold-based actuation.
- Integrated Blynk IoT platform for remote control and live telemetry dashboards via mobile.
- Implemented OTA firmware updates and NVS persistent storage, eliminating manual reconfiguration on restart.
- Awarded 3rd Place at GITAM Tech Exhibition 2025, competing against 200+ teams.

CodeLearn — Online Coding Platform — MongoDB · Node.js · Render

- Developed a full-stack coding platform featuring an in-browser code execution engine for multiple programming languages.
- Built separate teacher and student dashboards with performance tracking, submission history, and progress analytics.
- Deployed on Render with MongoDB Atlas backend; onboarded 15–20 active users in initial release.

Physics Lab Simulator with AI Chatbot — Python · Gemini API

- Created an interactive simulation platform for physics experiments, enabling hands-on learning without physical lab access.
- Integrated Google Gemini API as a contextual chatbot to answer student queries in real time within the simulator.

TECHNICAL SKILLS

Languages: Python, C, HTML, Java (Learning)
Computer Vision: OpenCV, MediaPipe
ML / AI Frameworks: PyTorch, YOLOv8, Stable-Baselines3, Gymnasium
AI / Tools: Gemini API, Agentic AI, GitHub Copilot, Claude AI, Google Colab
CS Fundamentals: Data Structures & Algorithms (basics)
IoT / Hardware: ESP32, Arduino, DHT11, Blynk IoT
Simulation: MATLAB, Simulink, Simscape
Database: MongoDB Atlas
Automation & Ops: Google Apps Script, Notion
Dev Tools: VS Code, Git, GitHub, Render

ACHIEVEMENTS

- 3rd Place — GITAM Tech Exhibition 2025 (200+ competing teams)
- Full Merit Scholarship — GITAM University, awarded for academic excellence at admission
- SGPA 10.0 / 10.0 — Semester 1, B.Tech CSE (AI & ML)
- SIH 2025 (Smart India Hackathon) — Participant
- Class Representative — CSE (AI & ML) batch, GITAM University

COURSES & TRAINING

Agentic AI Workshop — GITAM University	2025
Java & Data Structures — Self Study	In Progress